Physics of Complex Systems Homework 9

Jun Wei Tan*

Julius-Maximilians-Universität Würzburg

(Dated: July 1, 2026)

Problem 1 (Entanglement of a Pure 2-Qubit State)**.** Consider the 2-qubit state

$$\rho = |\psi\rangle\langle\psi|$$

with

$$|\psi\rangle = \frac{1}{5}|00\rangle + \frac{2i\sqrt{2}}{5}|01\rangle - \frac{4}{15}|10\rangle - \frac{8i\sqrt{2}}{15}|11\rangle.$$

(a) Use Mathematica or similar tools to compute the 4×4 matrix ρ in the qubit basis. Verify that ρ is pure.

(b) Compute the reduced density matrices $\rho^{(L)}$ and $\rho^{(R)}$ for the left and right qubit by carrying out the partial traces

$$\rho^{(L)}_{jk} = \sum_{m=0}^{1} \rho_{jm,km}, \qquad \rho^{(R)}_{mn} = \sum_{j=0}^{1} \rho_{jm,jn}.$$

(c) A pure state is called entangled if the reduced density matrices are mixed. Find out whether ρ is entangled or not.

Proof. (a) We define the ordered basis of the Hilbert space as $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ (in that order). Thus, the ket would be represented as

$$|\psi\rangle = \begin{pmatrix} \frac{1}{5} \\ \frac{2i\sqrt{2}}{5} \\ -\frac{4}{15} \\ -\frac{8i\sqrt{2}}{15} \end{pmatrix}.$$

We then compute the outer product in mathematica using

```
In[1]:=  v = 1/5, 2 I Sqrt[2]/5, -4/15, -8 I Sqrt[2]/15
```


* jun-wei.tan@stud-mail.uni-wuerzburg.de

```
In[2]:= ρ = Outer[Times, Conjugate[v], v]

Out[2]= {{1/25, (2 I Sqrt[2])/25, -(4/75), -((8 I Sqrt[2])/75)}, {-((2 I Sqrt[2])/25), 8/25, (8 I Sqrt[2])/75, -(32/75)}, {-(4/75), -((8 I Sqrt[2])/75), 16/225, (32 I Sqrt[2])/225}, {(8 I Sqrt[2])/75, -(32/75), -((32 I Sqrt[2])/225), 128/225}}
```

to find

$$\rho = \begin{pmatrix} \frac{1}{25} & \frac{2i\sqrt{2}}{25} & -\frac{4}{75} & -\frac{8i\sqrt{2}}{75} \\ -\frac{2i\sqrt{2}}{25} & \frac{8}{25} & \frac{8i\sqrt{2}}{75} & -\frac{32}{75} \\ -\frac{4}{75} & -\frac{8i\sqrt{2}}{75} & \frac{16}{225} & \frac{32i\sqrt{2}}{225} \\ \frac{8i\sqrt{2}}{75} & -\frac{32}{75} & -\frac{32i\sqrt{2}}{225} & \frac{128}{225} \end{pmatrix}.$$

We can square this matrix to get

$$\rho^2 = \begin{pmatrix} \frac{1}{25} & \frac{2i\sqrt{2}}{25} & -\frac{4}{75} & -\frac{8i\sqrt{2}}{75} \\ -\frac{2i\sqrt{2}}{25} & \frac{8}{25} & \frac{8i\sqrt{2}}{75} & -\frac{32}{75} \\ -\frac{4}{75} & -\frac{8i\sqrt{2}}{75} & \frac{16}{225} & \frac{32i\sqrt{2}}{225} \\ \frac{8i\sqrt{2}}{75} & -\frac{32}{75} & -\frac{32i\sqrt{2}}{225} & \frac{128}{225} \end{pmatrix}.$$

Since the matrix has not changed at all, we know that the trace remains the same, i.e. it is 1. We can also confirm this by evaluating

```
In[3]:= Tr[ρ.ρ]

Out[3]= 1
```

This is to be expected, since we had $\rho = |\psi\rangle\langle\psi|$ and hence

$$\rho^2 = |\psi\rangle \langle\psi|\psi\rangle \langle\psi| = |\psi\rangle\langle\psi|.$$

(b) In terms of our indexing, the multiindex (m, j) is mapped to the index $2(m-1)+j$, where we note that the indices m and j take values $\{0, 1\}$.

Thus, we can rewrite the reduced matrices as

$$\begin{aligned} \rho^{(L)}_{jk} &= \sum_{m=0}^{1} \rho_{jm,km} \\ &= \sum_{m=0}^{1} \rho_{(2(j-1)+m)(2(k-1)+m)} \end{aligned}$$

We can evaluate this with mathematica as

```
In[4]:= Table[Sum[ρ[[2 (j - 1) + m, 2 (k - 1) + m]], {m, 1, 2}], {j, 1, 2}, {k, 1, 2}]

Out[4]= {{9/25, -(12/25)}, {-(12/25), 16/25}}
```

We can also evaluate the right reduced density matrix as

$$\begin{aligned}\rho^{(R)}_{mn} &= \sum_{j=1}^{2} \rho_{jm,jn} \\ &= \sum_{j=1}^{2} \rho_{(2(j-1)+m)(2(j-1)+n)}\end{aligned}$$

We then evaluate this using

```
In[5]:= Table[Sum[ρ[[2 (j - 1) + m, 2 (j - 1) + n]], {j, 1, 2}], {m, 1, 2}, {n, 1, 2}]

Out[5]= {{1/9, (2 I Sqrt[2])/9}, {-((2 I Sqrt[2])/9), 8/9}}
```

In a better formatted matrix form, we have

$$\rho^{(L)} = \begin{pmatrix} \frac{9}{25} & -\frac{12}{25} \\ -\frac{12}{25} & \frac{16}{25} \end{pmatrix}, \qquad \rho^{(R)} = \begin{pmatrix} \frac{1}{9} & \frac{2i\sqrt{2}}{9} \\ -\frac{2i\sqrt{2}}{9} & \frac{8}{9} \end{pmatrix}.$$

(c) We can verify

$$(\rho^{(L)})^2 = \rho_L, \qquad (\rho^{(R)})^2 = \rho^{(R)}$$

using the following code

```
In[6]:= ρL.ρL

Out[6]= {{9/25, -(12/25)}, {-(12/25), 16/25}}

In[7]:= ρR.ρR

Out[7]= {{1/9, (2 I Sqrt[2])/9}, {-((2 I Sqrt[2])/9), 8/9}}
```

Since $\mathrm{tr}\left(\rho^{(L)}\right) = \mathrm{tr}\left(\rho^{(R)}\right) = 1$, we have $\mathrm{tr}\left((\rho^{(L)})^2\right) = \mathrm{tr}\left((\rho^{(R)})^2\right) = 1$, i.e. the states are pure. This means that the states are not entangled.

□

Problem 2 (Quantum Cryptography)**.** Encrypted communication requires that both parties share a key (a random sequence of classical bits) that can be used to encode and decode messages by bitwise XOR operations. The problem of encryption therefore reduces to safely generating identical keys on both sides.

Alice randomly generates a classical bit $a \in \{0, 1\}$, using it to prepare a qubit

$$|\psi\rangle = \begin{cases} |0\rangle, & a = 0, \\ |+\rangle, & a = 1, \end{cases}$$

where

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle).$$

She sends this qubit to Bob.

Likewise Bob randomly generates a classical bit $b \in \{0, 1\}$. He chooses one of the two measurements

$$\mathbf{M}_0 = |0\rangle\langle 0| - |1\rangle\langle 1|, \qquad \mathbf{M}_1 = |+\rangle\langle +| - |-\rangle\langle -|.$$

Bob measures the incoming qubit and communicates the measurement result $m = \pm 1$ publicly. They repeat the protocol until $m = -1$.

(a) Show that $a = 1 - b$ if $m = -1$.

(b) How many qubits are on average needed to generate one bit of the key?

(c) Suppose that Eve intercepts the communication in that she measures the transferred qubit by a projective measurement $\mathbf{E} = |\phi_+\rangle\langle\phi_+| - |\phi_-\rangle\langle\phi_-|$, where

$$|\phi_+\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\phi_-\rangle = \beta^*|0\rangle - \alpha^*|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

are two arbitrary orthogonal states. Show that Eve's interception generally changes the measurement statistics seen from the perspective of Bob and Alice, depending effectively on two parameters $u := |\alpha|^2$ and $v := \frac{1}{2}|\alpha + \beta|^2$.

(d) Determine u and v such that Bob and Alice cannot detect the interception.

(e) In the case of part (d), is it possible for Eve to obtain information about the key?

Proof. (a) Suppose $a = 0$. Thus Alice sends the state $|0\rangle$ to Bob. If Bob chooses M_0, the chance of measuring $m = -1$ is given by

$$\mathbb{P}(m = -1 | M_0, |0\rangle) = | \langle 1|0\rangle |^2 = 0.$$

Thus, if Bob chooses M_0, there is no chance of measuring -1. This means that Bob must have chosen M_1, i.e. $b = 1$.

Now suppose $a = 1$, and Alice sends the state $|1\rangle$ to Bob. If Bob has chosen the apparatus M_1, the probability of a measurement yielding -1 is given by

$$\mathbb{P}(m = -1 | M_1, |+\rangle) = | \langle -|+\rangle |^2 = 0.$$

Thus, if Bob has measured a -1, he must have chosen the apparatus M_0. We are then forced to conclude $b = 0$.

In any case, $a = 1 - b$.

For the sake of completeness, we compute the probability of measuring $m = -1$ when using the "correct" measurement apparatus:

$$\mathbb{P}(m = -1 | M_1, |0\rangle) = | \langle -|0\rangle |^2 = \frac{1}{2}$$
$$\mathbb{P}(m = -1 | M_0, |+\rangle) = | \langle 1|+\rangle |^2 = \frac{1}{2}$$

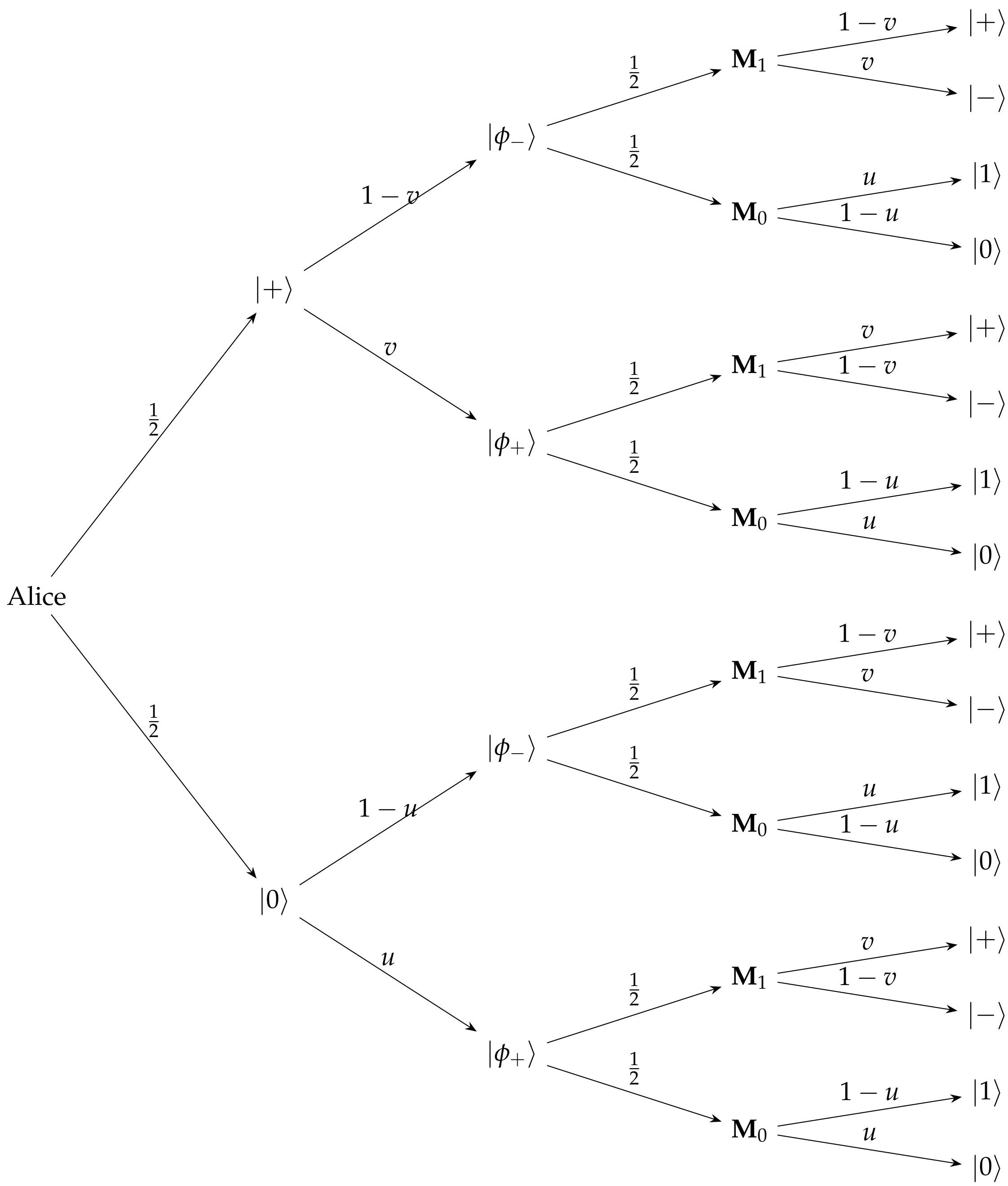
(b) As we proved in the previous part, the probability of measuring $m = -1$ is $\frac{1}{2}$ given that we use the "correct" apparatus (and 0 otherwise). Since the probability that we use the "correct" apparatus is $\frac{1}{2}$, we have, independently of the value of a,

$$\mathbb{P}(m = -1) = \mathbb{P}(\text{correct apparatus}) \cdot \frac{1}{2} = \frac{1}{4}.$$

We thus need 4 qubits on average to generate one bit of the key.

(c) We compute the necessary inner products:

$$\begin{aligned}
\langle 0|\phi_+\rangle &= \alpha, & | \langle 0|\phi_+\rangle |^2 &= |\alpha|^2 = u \\
\langle 1|\phi_+\rangle &= \beta, & | \langle 1|\phi_+\rangle |^2 &= |\beta|^2 = 1 - u \\
\langle 0|\phi_-\rangle &= \beta^*, & | \langle 0|\phi_-\rangle |^2 &= |\beta|^2 = 1 - u \\
\langle 1|\phi_-\rangle &= -\alpha^*, & | \langle 1|\phi_-\rangle |^2 &= |\alpha|^2 = u
\end{aligned}$$



$$\langle -|\phi_+\rangle = \frac{1}{\sqrt{2}}(\alpha - \beta), \qquad |\langle -|\phi_+\rangle|^2 = \frac{1}{2}|\alpha - \beta|^2 = 1 - v$$

$$\langle +|\phi_+\rangle = \frac{1}{\sqrt{2}}(\alpha + \beta) \qquad |\langle +|\phi_+\rangle|^2 = \frac{1}{2}|\alpha + \beta|^2 = v$$

$$\langle -|\phi_-\rangle = \frac{1}{\sqrt{2}}(\beta^* + \alpha^*) \qquad |\langle -|\phi_-\rangle| = \frac{1}{2}|\beta^* + \alpha^*|^2 = v$$

$$\langle +|\phi_-\rangle = \frac{1}{\sqrt{2}}(\beta^* - \alpha^*) \qquad |\langle +|\phi_-\rangle|^2 = \frac{1}{2}|\beta^* - \alpha^*|^2 = 1 - v$$

where we have used the identity

$$2v = |\alpha + \beta|^2 = |\alpha|^2 + |\beta|^2 + 2\,\mathrm{Re}(\alpha^* \beta)$$
$$|\alpha - \beta|^2 = |\alpha|^2 + |\beta|^2 - 2\,\mathrm{Re}(\alpha \beta)$$

and hence

$$|\alpha + \beta|^2 + |\alpha - \beta|^2 = 2,$$

leading to

$$\frac{1}{2}|\alpha - \beta|^2 = 1 - v$$

Now, we go down the various probability branches as shown in the figure below to find the statistics

a b	$\mathbb{P}(m = 1)$	$\mathbb{P}(m = -1)$
0 0	$u^2 + (1-u)^2$	$2u(1-u)$
0 1	$uv + (1-u)(1-v)$	$u(1-v) + v(1-u)$
1 0	$vu + (1-v)(1-u)$	$v(1-u) + u(1-v)$
1 1	$v^2 + (1-v)^2$	$2v(1-v)$

It is clear that the statistics are changed: For example, it is clear that we can choose u and v such that the probabilities $\mathbb{P}(m = -1)$ are all nonzero, which means there is a chance of measuring $m = -1$ for all configurations; this contradicts (a).

(d) This depends largely on how "undetectable" is defined. We will take it to mean the result in part (a), i.e. that if the measurement m yields the result -1, then $b = 1 - a$. This means that the probabilities on the right in the first and last row have to vanish.

This means that $u = 0$ or 1, and $v = 0$ or 1. Note that some of these possibilities are generically not possible due to the constraint $|\alpha|^2 + |\beta|^2 = 1$. For example, $u = v = 0$ would imply $\alpha = \beta = 0$, which is not possible. It is also not possible to have $u = 1$ and $v = 1$, or $u = 1$ and $v = 0$, because $u = 1$ implies $|\alpha| = 1$, and thus $\beta = 0$.

Thus, the only possibility is $u = 1$ and $v = 0$, implying $|\alpha| = 1$ and $\beta = 0$. Since the phases cancel, we can just declare $\alpha = 1$.

(e) The measurements Eve does are just the measurements in the computational basis $|0\rangle$ and $|1\rangle$. Eve cannot obtain complete information about the key. If Eve measures the key $|1\rangle$, then she can tell that the state Alice prepared was in the state $|+\rangle$; however, if she measures the ket $|-\rangle$, then she cannot say about what bit Alice has.

This follows from the nonorthogonality of the states Alice has prepared, which is why it is required for the protocol. $\square$